

QIJIE HE

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RESEARCH INTERESTS

Generative AI; Operations Research; Transportation Safety

EDUCATION

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| Southeast University , Nanjing, China | Sep. 2023 – Jun. 2026 |
| M.Eng. in Transportation Engineering | Supervisor: Prof. Ziyuan Pu |
| • Dissertation: Safety-Critical Static Road Scenario Generation Considering Highway Alignment Risk Factors | |
| Monash University , Melbourne, Australia | Sep. 2023 – Jun. 2026 |
| M.S. in Transportation Systems (Joint Program with Southeast University) | Supervisor: Dr. Zhuo Chen |
| Foshan University , Foshan, China | Sep. 2016 – Jun. 2020 |
| B.Eng. in Transportation Engineering | |

PUBLICATIONS

(*co-first author, †corresponding author)

Refereed Journal Articles

- J1. Chen J*, **He Q***, Liu P, Ma W, Zheng N, Pan Liu, Pu Z†. (2026) “Traffic Crash Data Augmentation with Multi-type Variables using Hybrid VAE-Diffusion Generative Neural Networks for Enhancing Crash Frequency Modeling.” *Accident Analysis and Prevention*. 231: 108508. [\[Link\]](#)
- J2. Zhao H, Wang S, **He Q**, Pu Z†, Ren Y, Yu H. (2026) “CAD-SGLLM: A Context-Aware Decoding Powered Large Language Model for Safety-Critical Scenario Generation.” *Transportmetrica A: Transport Science* (Early Access). [\[Link\]](#)

Manuscripts Under Review

- M1. Wang S, **He Q**, Zhao H, Chen J, Pu Z†. “Stable Generation of Safety-Critical Traffic Scenarios via a Multi-Dimensional Prior Knowledge-Enhanced Generative Adversarial Network.” *Submitted to Journal of Intelligent and Connected Vehicles*.

Conference Poster & Oral Presentation

- P1. **He Q**, Wang Y, Chen Z†. (2026) “Exploring the Impact of Active Transport Amenities on Lifestyle-Related Health Risks: A Partial Least Squares Structural Equation Modeling Analysis of Diabetes Incidence.” Paper Number TRBAM-26-05660. Poster. *Transportation Research Board 105th Annual Meeting*, 11–15 January, Washington, D.C.
- P2. **He Q**, Chen J, Liu P, Ma W, Pu Z†. (2025) “Enhancing Crash Frequency Modeling with Mixed-Type Data Using a Hybrid VAE-Diffusion-Based Generative Approach.” Paper Number TRBAM-25-06171. Poster. *Transportation Research Board 104th Annual Meeting*, 5–9 January, Washington, D.C.

Refereed Conference Proceedings

- C1. Chen Z, **He Q**, Fan D, Luo Z. (2019) “An Intelligent Bus Scheduling System Based on Real-Time Passenger Flow Data.” *Proceedings from the International Conference on Robots & Intelligent System (ICRIS)*, 15–16 June, Haikou, China, 174–178. [[Link](#)]

RESEARCH EXPERIENCE

Department of Civil and Environmental Engineering, Monash University

Suzhou, China

Research Assistant/Teaching Assistant, Supervisor: Dr. Zhuo Chen

Aug. 2026 – present

1. **Identifying Critical Charging Corridors for Electric Trucks: A Simulation-Optimization Approach with LLM-Based Generative Agents**
 - **Reviewing** literature on LLM-based generative agents and generative agent-based modeling (GABM) and **identifying** research gaps to refine the study scope.
2. **Sustainable Vertiport Location Allocation Under Dynamic Demand: A Unified Graph Diffusion Model and Reinforcement Learning Approach**
 - **Mentoring** a student researcher in revising the methodology and experimental design; **co-authored** a manuscript submitted to the TRB 106th Annual Meeting.

Graduate Student, Supervisor: Dr. Zhuo Chen

Mar. 2024 – Jun. 2026

3. **Transportation and Health: Exploring the Impact of Active Transport Amenities on Health Risks**
 - **Collected** data from multiple sources (POI, demographics, and survey data), and **processed** it using QGIS and Python.
 - **Reviewed** research articles, **designed** experiments, and **conducted** Partial Least Squares Structural Equation Modeling Analysis using SmartPLS 4.
 - **Authored** a paper that was accepted for poster presentation at the TRB 105th Annual Meeting (see **P1**).

School of Transportation, Southeast University

Nanjing, China

Graduate Student, Supervisor: Prof. Ziyuan Pu

Sep. 2023 – Jun. 2026

1. **Master's Dissertation: Safety-Critical Static Road Scenario Generation Considering Highway Alignment Risk Factors**
 - **Analyzed** crash report data through crash frequency modeling and **quantified** the effects of highway alignment risk factors (e.g., horizontal curve radius, longitudinal grade) on crash risk.
 - **Developed** a risk-conditioned generative model to generate highway road segments with controllable risk levels for simulation-based autonomous vehicle testing, and **evaluated** the generated scenarios in CARLA.
 - **Co-authored** two related papers on safety-critical scenario generation (see **J2**, **M1**), contributing to idea formulation and manuscript revision.
2. **Generative AI-based Imbalanced Data Augmentation and Data-Driven Crash Frequency Modeling**
 - **Curated** crash data using Python, and **reviewed** research articles on ML-based crash frequency prediction and generative AI.
 - **Tested** generative AI methods using the PyTorch framework, **developed** ML-based crash frequency prediction models, and interpreted the results via SHAP.
 - **Authored** and submitted a paper, and **presented** it as a poster at the TRB 104th Annual Meeting (see **P2**).
 - **Extended** the work into a journal article as co-first author, leading the additional comparative experiments, crash data analysis, manuscript writing, and peer-review revisions through acceptance in *Accident Analysis and Prevention* (see **J1**).

School of Transportation, Civil Engineering & Architecture, Foshan University

Foshan, China

Undergraduate Student, with Dr. Cong Zhai

2019

1. **“Challenge Cup” Student Competition: Bus Scheduling System Optimization**

- **Reviewed** research articles, **analyzed** the operational problem, **defined** modeling assumptions, decision variables, objective function, and model constraints.
- **Co-authored** and submitted a paper that was accepted to a conference proceeding (see **C1**).

AWARDS AND SCHOLARSHIPS

Suzhou Alumni Scholarship, Southeast University Education Foundation	2025
Third Prize, "HUAWEI Cup" The 21 st China Post-Graduate Mathematical Contest in Modeling	2024
Honorable Mention, The 12 th National Competition of Transport Science and Technology for Students	2017

TECHNICAL SKILLS

Programming: Python, R, LaTeX
Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas
Tools & Simulation: QGIS, SmartPLS, CARLA

EMPLOYMENT HISTORY

Shenzhen Urban Transport Planning Center Co., Ltd., Shenzhen, China <i>Junior Engineer, Dongguan Branch</i>	Jun. 2020 – May 2021
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